

My white board writing from week 5 - 22nd August

I went through the slides related to lecture 5. They are very detailed (including the numerical examples) and I abstain from reproducing them here.

However, I will reproduce in details the discussion related to copulae.

I first reminded the fact that for a random variable X with a continuous cumulative distribution function $F_X(\cdot)$, the random variable $Y = F_X(X)$ is uniformly distributed in $[0, 1]$ (this is the statement called the Probability Integral Transform).

I shortly discussed the proof of Sklar's Theorem (for $p=2$).

We need to show claims in two directions:

→ Let $F(x_1, x_2)$ is the joint distribution of (X_1, X_2) and let us denote the resulting marginal distributions as $F_{X_1}(\cdot)$ and $F_{X_2}(\cdot)$. Because of the monotonicity of the cdf we can write:

$$F(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2) = P(F_{X_1}(X_1) \leq F_{X_1}(x_1), F_{X_2}(X_2) \leq F_{X_2}(x_2))$$

$$=: C(y_1, y_2) \quad \text{with } y_1 = F_{X_1}(x_1) \text{ and } y_2 = F_{X_2}(x_2),$$

hence the copula related to $F(x_1, x_2)$ is defined: namely: for each (x_1, x_2) we can calculate $F_{X_1}(x_1) = y_1$, $F_{X_2}(x_2) = y_2$ and the value of the copula is just given by $P(F_{X_1}(X_1) \leq y_1, F_{X_2}(X_2) \leq y_2)$ (which is the same as the cdf value $F(x_1, x_2)$).

← Similarly, applying the inverse of the marginal cdf's, we can write for a give copula at $y_1 \in [0, 1], y_2 \in [0, 1]$ that:

$$C(y_1, y_2) = P(X_1 \leq F_{X_1}^{-1}(y_1), X_2 \leq F_{X_2}^{-1}(y_2))$$

and denoting $x_1 = F_{X_1}^{-1}(y_1)$, $x_2 = F_{X_2}^{-1}(y_2)$, we

can attach to this $P(X_1 \leq x_1, X_2 \leq x_2)$ which is then the value of $F(x_1, x_2)$.

Then I stressed on the importance of the relation between joint distributions and copulas. The relation becomes very clear when we consider the densities. From $F(x_1, x_2) = C(F_{X_1}(x_1), F_{X_2}(x_2))$, by taking mixed partial derivatives w.r. to x_1, x_2 , i.e.:

$$\frac{\partial^2}{\partial x_1 \partial x_2} F(x_1, x_2) = f(x_1, x_2) = c(F_{X_1}(x_1), F_{X_2}(x_2)) f_{X_1}(x_1) f_{X_2}(x_2) (*)$$

where $c(u, v) = \frac{\partial^2}{\partial u \partial v} C(u, v)$ is used as notation.

The function $c(u, v)$ is the copula density. The relation (*) clearly shows that the contribution to the joint density $f(x_1, x_2)$ of X_1 and X_2 comes from two parts: one that is "responsible" for the dependence ($c(u, v) = \frac{\partial^2}{\partial u \partial v} C(u, v)$), and another one that takes into account the marginal information $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ only. It also shows why $C(u, v) = uv$ is called the independence copula (since then $c(u, v) = 1$) and we get $f(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$ (as it should be if X_1 and X_2 were independent. So: the independence copula density is $\equiv 1$).

I then discussed some special copulas like the Gaussian, the Gumbel - Hongard, the class of Archimedean copulas. The Gumbel - Hongard copula is parametrized by one parameter θ : $C_\theta(u_1, u_2, \dots, u_p) = \exp\left\{-\left[\sum_{j=1}^p (-\log u_j)^\theta\right]^{\frac{1}{\theta}}\right\}$

It is easy to see that

if $\theta=1$ we get the independence copula. In general, θ can vary in $[1, \infty)$ and governs the strength of the dependence. When $\theta \rightarrow \infty$ we get the so called upper bound copula $\min(u_1, u_2, \dots, u_p)$ in a limit.

Indeed since $-\log u_j = a_j$ gives a set of positive a_j values, if we denote by a_{j^*} the maximal of them (which is obtained when u_{j^*} is minimal) then we have $\left(\sum_{j=1}^p a_j^\theta\right)^{\frac{1}{\theta}} = a_{j^*}^\theta \left(\underbrace{1 + c}_{\text{finite value since } c \leq p}\right)^{\frac{1}{\theta}}$ $\theta \rightarrow \infty \Rightarrow a_{j^*}^\theta = -\log u_{j^*}$

Hence in a limit, as $\theta \rightarrow \infty$ we get

$$\exp(-(-\log u_{j^*}^\theta)) = \exp(\log u_{j^*}^\theta) = u_{j^*}^\theta = \min(u_1, \dots, u_p)$$

In the definition of Archimedean copulas, the generator $\Phi(\cdot)$  was important. It is a

continuous strictly decreasing function on $[0, 1]$ such that $\Phi(1) = 0$. For given generator Φ ,

$$C(u_1, u_2, \dots, u_p) = \Phi^{-1}(\Phi(u_1) + \dots + \Phi(u_p))$$

gives the copula. The Gumbel - Hongard copula is an Archimedean copula with a generator $\phi(t) = (-\log t)^\theta$.

Indeed: Set $\phi(t) = y \Rightarrow -y^{\frac{1}{\theta}} = \log t \Rightarrow t = e^{-y^{\frac{1}{\theta}}} = \phi^{-1}(y)$

But then $\phi(u_1) + \phi(u_2) + \dots + \phi(u_p) = \sum_{i=1}^p (-\log u_i)^\theta$ and

$$\boxed{\Phi^{-1}[\phi(u_1) + \dots + \phi(u_p)] = e^{-\left[\sum_{j=1}^p (-\log u_j)^\theta\right]^{\frac{1}{\theta}}} \text{ qed.}}$$